



Cardinality constrained multi-period mean–variance portfolio optimization with regime-switching parameters[☆]

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ABSTRACT

This paper studies the cardinality-constrained multi-period mean–variance (CMMV) portfolio optimization problem in markets with the Markov regime-switching parameters. Given practical trading considerations, investors typically select a subset of assets from a larger asset pool to construct their portfolios, known as cardinality-constrained portfolios. Due to the combinatorial nature of the cardinality constraint, developing a feedback-type portfolio policy for multi-period portfolio optimization poses a significant challenge. We address this challenge for such a problem by leveraging the special structure of cardinality-constrained quadratic optimization (CCQO), which allows us to develop the wealth separation property. Through this property and the utilization of embedded methods, we derive the analytical form of the dynamic portfolio policy and the expression of the MV efficient frontier via dynamic programming. We demonstrate that portfolio weights can be recursively computed by solving a sequence of CCQO sub-problems for each type of market state. This solution accommodates several market settings as its special cases, such as markets with stationary distributions or with independent returns. Utilizing numerical examples calibrated from real market data, we showcase that the dynamic CMMV portfolio policy outperforms various benchmark portfolios, including static cardinality-constrained portfolios and dynamic ones that disregard return correlations.

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1. Introduction

The mean–variance (MV) portfolio selection model, initially proposed by Markowitz (1952), serves as a fundamental tool for constructing diversified portfolios widely used in practical applications. Among various extensions of Markowitz's one-period MV portfolio selection models (Kalayci, Ertenlice, & Akbay, 2019; Kolm & Tütüncü, 2014), one line of research endeavors to develop feedback-type portfolio policies for the multi-period mean–variance (MMV) dynamic portfolio optimization models. This is a nontrivial task, as the variance term introduces non-separability in the context of dynamic programming. Seminal works (Li & Ng, 2000) and Zhou and Li (2000) overcome this challenge by

embedding the problem into a parameterized linear–quadratic optimal control problem and derive analytical portfolio policies (see Cui, Gao, Li, and Shi (2022) for a brief survey). In line with this, the dynamic MV portfolio decision model has been explored across various market settings, encompassing models with independent returns (Cui, Gao, Li, & Li, 2014; Li & Ng, 2000), mean-reverting returns (Chiu & Wong, 2011), and parameters subject to Markov regime switching (Çakmak & Özekici, 2006; Costa & Araujo, 2008; Yin & Zhou, 2004). Conversely, considering practical implementation, much research (see Mencarelli and D'Ambrosio (2019)) focuses on addressing challenges posed by various practical trading restrictions in single-period (static) portfolio optimization, including cardinality constraints (Bertsimas & Cory-Wright, 2022; Gao & Li, 2013). However, existing research provides limited insights into multi-period MV portfolio optimization when cardinality constraints are involved, and this study aims to fill this gap.

In this paper, we study the cardinality-constrained multi-period mean–variance (CMMV) portfolio optimization problem within a market characterized by Markov regime-switching parameters. Leveraging the solution structure of cardinality-constrained quadratic optimization (CCQO), we unveil the wealth

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separation property of the value function. This discovery enables us to devise an analytical portfolio policy using dynamic programming. We demonstrate that the allocation vector, which dictates portfolio weights, can be precomputed offline by solving a sequence of CCQO problems for each market state. This property facilitates the application of well-established numerical solution methods (Gao & Li, 2013)(Bertsimas & Cory-Wright, 2022) for the notoriously NP-hard CCQO problem. Our solution scheme involves the computation of a sequence of variables $\{\theta_t(s_t)\}_{t=0}^T$ defined for each market state s_t . Such variables have precise financial meanings which directly link to the conditional Sharpe ratio for a given market state. This analytical solution reveals how the market condition and cardinality size affect the portfolio policy. Besides the solutions for the general market setting, we delineate how to customize the solution for specific markets, such as those with stationary (time-invariant) parameters and dependent returns. Furthermore, through numerical experiments, we compare our CMMV portfolio policy with several benchmark policies. Empirical results showcase that our approach surpasses the benchmark policies in terms of the Sharpe ratio. These findings underscore the benefits of employing a multi-period feedback MV portfolio policy in portfolio management, particularly in the presence of cardinality constraints and stochastic market parameters.

Our work advances the current development of various MMV portfolio optimization models, particularly those involving portfolio constraints (see, e.g., Cui et al. (2022) for a brief review). For developing the feedback portfolio policy, Cui et al. (2014), Cui, Li, and Li (2017), and Wu, Gao, Li, and Shi (2018) have developed semi-analytical solutions for MMV models with no-shorting and cone constraints. Zhu, Li, and Wang (2004) and Costa and Araujo (2008) address the problem with a no-bankruptcy constraint. Several studies incorporate transaction costs into the dynamic MV portfolio model. For example, Dai, Xu, and Zhou (2010) develops solutions using variational inequality under a continuous-time model. Pun and Ye (2022) carefully models transaction costs by distinguishing between risky and risk-free accounts for the MMV portfolio model, providing a semi-analytical solution using the embedding method. Cui, Gao, and Shi (2019) integrates the management fee into the MMV portfolio optimization model and obtains a semi-analytical solution. For cardinality-constrained MV portfolio optimization, the single-period models focus on developing efficient numerical algorithms for this NP-hard problem (see Bertsimas and Cory-Wright (2022), Tillmann, Bienstock, Lodi, and Schwartz (2022) and Gao and Li (2013)). Some research extends this model to a multi-period setting by developing numerical solutions for the open-loop portfolio policy (see Liu, Zhang, and Wang (2016) and Wu, Xie, Ge, and De Simone (2024)). While our work aims to develop a type of feedback portfolio policy called the pre-committed policy for the MMV problem (see the brief review in Cui et al. (2022)), another branch of research studies the time-consistent policy for MMV problems (see Basak and Chabakauri (2010), Björk and Murgoci (2014), Wu (2013) and Wang and Wei (2019)). Note that if the investor fixes the investment horizon and adheres to the pre-committed policy, such a policy will outperform the time-consistent policy (see Lemma 4.4 in van Staden, Dang, and Forsyth (2021) and Vigna (2020)).

Our work is also related to the MMV portfolio in the market with random coefficients. The Markov regime switch-based market model is popular as it captures the serial correlation of market states by a tractable structure. It has been incorporated into MMV portfolio optimization (see Çakmak and Özekici (2006), Celikyurt and Özekici (2007), Chen and Yang (2011), Costa and Araujo (2008), Wu and Li (2012) and Zhang, Jin, Wei, and Yin (2022)) and other types of portfolio decision models (see Costa

and Kwon (2019), Pun, Wang, and Yan (2023) and Zhu, Xie, Ching, and Siu (2016)). As for the market model with random coefficients, Lim and Zhou (2002) and Shen (2015) study the dynamic MV portfolio optimization under such a setting. Gao, Li, Cui, and Wang (2015) studies the time-cardinality-constrained MMV portfolio in the market with serially correlated returns, where the time-cardinality constraint aims to select a subset of periods to invest in risky assets, differing from the current research, which aims to select a subset of risky assets from an asset pool.

Notations: We employ boldface letters to represent vectors or matrices. The notation $\mathbf{e}$ denotes the all-one vector of appropriate dimension. The expression $\mathbf{A} \succ 0$ indicates that $\mathbf{A}$ is a positive definite matrix, and $\mathbb{S}_+^n$ denotes the set of all $n \times n$ positive definite matrices. Given a vector $\mathbf{x} \in \mathbb{R}^n$, the notation $\text{diag}(\mathbf{x}) \in \mathbb{R}^{n \times n}$ denotes the diagonal matrix with elements of the diagonal being $\mathbf{x}$. The symbol $\mathbf{A}^\dagger$ denotes the Moore–Penrose pseudoinverse of the matrix $\mathbf{A}$ (Horn & Johnson, 2012). We use $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ to denote the multivariate normal distribution with $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ representing the mean and covariance, respectively. For a problem $\mathcal{P}$, $v(\mathcal{P})$ denotes its optimal objective value. Given vector $\mathbf{x} = (x(1), \dots, x(n))^\top$, the l_0 -norm is defined as $\|\mathbf{x}\|_0 = \sum_{i=1}^n |\text{sign}(x(i))|$, where $\text{sign}(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$ is the sign function, i.e., $\text{sign}(a) = 1$ if $a > 0$, $\text{sign}(a) = -1$ if $a < 0$, and $\text{sign}(a) = 0$ if $a = 0$.

2. Market model and problem formulation

The market consists of one risk-free asset and n risky assets whose prices evolve over $T + 1$ discrete time instances, labeled as $0, \dots, T$. We use $S_t(i)$ to denote the price of the i th risky asset at time t and $\mathbf{R}_t = (R_t(1), \dots, R_t(n))^\top$ to denote the random return vector of these n risky assets, where $R_t(i) = S_t(i)/S_{t-1}(i)$ for $i = 1, \dots, n$ and $t = 1, \dots, T$. The risk-free asset's return of the t th period is r_t , which is assumed to be deterministic for $t = 1, \dots, T$.² All the random variables/processes in this paper are defined on a complete probability space $\{\Omega, \{\mathcal{F}_t\}_{t=0}^T, \mathbb{P}\}$, where $\mathbb{P}$ is the probability measure, $\{\mathcal{F}_t\}_{t=0}^T$ is the information set satisfying $\mathcal{F}_0 = \{\emptyset, \Omega\}$, and $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$.³ In this setting, $\mathbf{R}_t$ is $\mathcal{F}_t$ -measurable, i.e., it can be observed after time t . For simplicity, we use $E_t[\cdot] \triangleq E[\cdot | \mathcal{F}_t]$ to denote the conditional expectation of some random variables with respect to $\mathcal{F}_t$. At time $t = 0$, $E_0[\cdot] = E[\cdot | \mathcal{F}_0]$ represents the unconditional expectation.

We adopt the standard framework of the Markov regime-switching market (Çakmak & Özekici, 2006; Costa & Araujo, 2008; Wu et al., 2018) by assuming that the expected value and covariance matrix of the risky assets' return $\{\mathbf{R}_t\}_{t=1}^T$ satisfy a finite-state Markov chain model as follows.

Assumption 1. The market state $s_t \in \mathbb{M} \triangleq \{1, \dots, m\}$ follows a finite-state Markov chain model for $t = 0, \dots, T$. The state transition probability $p_t(i, j) \triangleq \mathbb{P}(s_{t+1} = j | s_t = i)$ represents the probability of transitioning from state i to state j for all $i, j \in \mathbb{M}$ and $t = 0, 1, \dots, T - 1$. At time $t = 0$, the initial state is given by s_0 , and the subsequent states s_t , $t = 1, \dots, T$, transition according to $\{p_t(i, j)\}_{i, j \in \mathbb{M}}$. In this market, for a given market state $s_t = i \in \mathbb{M}$ at time t , the expected value and covariance matrix of $\mathbf{R}_{t+1}$ are $\boldsymbol{\mu}_t(i) \triangleq E_t[\mathbf{R}_{t+1} | i]$ and $\boldsymbol{\Sigma}_t(i) \triangleq \text{Cov}_t[\mathbf{R}_{t+1} | i] = E_t[\mathbf{R}_{t+1} \mathbf{R}_{t+1}^\top | i] - \boldsymbol{\mu}_t(i) \boldsymbol{\mu}_t(i)^\top$, respectively, for $t = 0, \dots, T - 1$.

² This simplification allows us to isolate the impact of regime-switching in risky assets, is reasonable for the short to medium-term horizons considered, and enhances model tractability.

³ Our market model follows a similar setting as introduced by Föllmer and Schied (2016), where a rigorous discrete-time market model is defined. The information set $\{\mathcal{F}_t\}_{t=0}^T$ is also referred to as the *filtration*, with $\mathcal{F}_t$ representing the σ -algebra that models the class of all events observable up to time t .

Assumption 1 establishes a comprehensive model for assets' returns with Markov chain-based parameters. A specific instance within this framework is the time-variant Gaussian type of returns, represented as $\mathbf{R}_{t+1} = \boldsymbol{\mu}_t(s_t) + \boldsymbol{\Sigma}_t(s_t)\mathbf{w}_t$ for $t = 0, \dots, T-1$, where s_t follows Markov chain model and $\{\mathbf{w}_t \in \mathbb{R}^n\}_{t=0}^{T-1}$ is a sequence of time-independent Gaussian random vectors with zero mean and identity covariance matrix. We assume that the random noises $\mathbf{w}_t$ and the Markov state process s_t are mutually independent and that the Markov state s_t is observable to investors. Note that the parameters $\{\boldsymbol{\mu}_t(i), \boldsymbol{\Sigma}_t(i), i \in \mathbb{M}\}_{t=1}^T$ can be calibrated using statistical methods, such as those outlined by Pun et al. (2023), leveraging historical data. In this work, we also assume the following non-degenerating condition holds,

$$\boldsymbol{\Sigma}_t(i) \succ 0, \quad i \in \mathbb{M}, \quad t = 1, \dots, T. \quad (1)$$

The above condition for $\boldsymbol{\Sigma}_t(i)$ is a natural extension of the non-degenerate condition commonly imposed in MMV portfolio model with independent returns (see, e.g., Li and Ng (2000) and Cui et al. (2014)).

An investor enters the market with an initial wealth $x_0 > 0$ and re-balances the portfolio at the beginning of each period. We use $\mathbf{u}_t = (u_t(1), \dots, u_t(n))^T$ to represent the portfolio decision where $u_t(i)$ is the wealth invested in the i th risky asset at time t for $t = 0, 1, \dots, T-1$. We require the time- t portfolio policy to be $\mathcal{F}_t$ -measurable, i.e., the policy $\mathbf{u}_t$ only depends on the historical information generated up to time t . Motivated by the necessity for the investor to select a subset of risky assets from the asset pool to construct the portfolio,⁴ we introduce the following cardinality constraint for a given positive integer q with $0 < q \leq n$ (see, e.g., Gao and Li (2013) and Bertsimas and Cory-Wright (2022)),

$$\mathbb{U}(q) \triangleq \{\mathbf{u} = (u(1), \dots, u(n))^T \mid \|\mathbf{u}\|_0 \leq q\}, \quad (2)$$

The constraint in (2) implies that the nonzero elements in $\mathbf{u}$ should not exceed q . Then, the constraint $\mathbf{u}_t \in \mathbb{U}(q)$ means that the investor selects at most q active assets from total n assets to construct the portfolio $\mathbf{u}_t$.

Additionally, for any portfolio $\mathbf{u} = (u(1), \dots, u(n))^T$, we introduce the indicator operator as

$$\mathcal{I}(\mathbf{u}) \triangleq \{ (z(1), \dots, z(n))^T \in \{0, 1\}^n \mid z(i) = 1 \text{ if } u(i) \neq 0; \\ z(i) = 0, \text{ if } u(i) = 0, \forall i \},$$

which returns a binary vector indicating the active elements of $\mathbf{u}$. If $\mathbf{u} \in \mathbb{U}$, we then define the feasible set for $\mathcal{I}(\mathbf{u}) = \mathbf{z}$ as follows,

$$\mathbb{B}(q) \triangleq \{\mathbf{z} \in \{0, 1\}^n \mid \sum_{i=1}^n z(i) \leq q\}.$$

For example, if $n = 4$ and $q = 2$, the above set becomes $\mathbb{B}(q) = \{(1, 0, 0, 1)^T, (1, 0, 1, 0)^T, \dots\}$, which indicates all possible active portfolios. Clearly, the restriction of the active index set, $\mathcal{I}(\mathbf{u}) \in \mathbb{B}(q)$, is equivalent to $\mathbf{u} \in \mathbb{U}(q)$. From the set $\mathbb{B}(q)$, we may understand the combinatorial nature of the constraint $\mathbf{u} \in \mathbb{U}(q)$, i.e., the cardinality of the set $\mathbb{B}(q)$ is $\sum_{k=1}^q C_n^k$,⁵ which is a large number when n and q are large.

It is important to note that the cardinality constraint defined in (2), along with constraints like no-shorting constraint ($\mathbf{u}_t \geq \mathbf{0}$) and no-borrowing constraints (such as $\mathbf{1}^T \mathbf{u}_t \geq 0$ or even more generally $\mathbf{A}^T \mathbf{u}_t \geq 0$ for a matrix $\mathbf{A}$ that defines restrictions on borrowing for certain portfolio subsets), fall under the category of

cone constraints. A key characteristic of cone constraints is that if a portfolio $\mathbf{u}$ is feasible, then any positively scaled portfolio $\alpha \mathbf{u}$ ($\alpha > 0$) is also feasible. This property is crucial for the dynamic programming approach used in this paper to derive analytical solutions. Moreover, due to this property, our methodology can be naturally extended to other cone constraints. In contrast, for non-conic constraints—such as transaction costs, threshold constraints, or upper bounds on portfolio weights—we recommend alternative numerical methods, such as those proposed in Cui et al. (2022), Lobo, Fazel, and Boyd (2007) and Li, Uysal, and Mulvey (2022).

We denote the investor's wealth level as x_t at time t for $t = 0, \dots, T-1$. Given portfolio $\mathbf{u}_t$ at time t , the allocation to the riskless asset is $x_t - \sum_{i=1}^n u_t(i)$. Thus, the dynamics of the wealth process satisfies the following equation,

$$x_{t+1} = r_{t+1}x_t + \mathbf{P}_{t+1}^T \mathbf{u}_t, \quad t = 0, \dots, T-1, \quad (3)$$

where $\mathbf{P}_{t+1} \triangleq \mathbf{R}_{t+1} - r_{t+1}\mathbf{e}$ denotes the excess return vector for $t = 0, \dots, T-1$. We use $\text{Var}_t[x] = E_t[x^2] - (E_t[x])^2$ to denote the conditional variance of the random variable x given filtration $\mathcal{F}_t$. Then, we introduce the following cardinality-constrained multi-period MV portfolio optimization (CMMV) problem,

$$\begin{aligned} \mathcal{P}_{\text{cmmv}}^1(\omega, q) : \quad & \min_{\{\mathbf{u}_t\}_{t=0}^{T-1}} \text{Var}_0[x_T] - 2\omega E_0[x_T] \\ & \text{s.t. } x_t \text{ and } \mathbf{u}_t \text{ satisfy (3),} \\ & \mathbf{u}_t \in \mathbb{U}(q), \quad t = 0, \dots, T-1, \end{aligned} \quad (4)$$

where $\omega \geq 0$ is the weighting parameter balancing the importance of the expected value $E_0[x_T]$ and the variance $\text{Var}_0[x_T]$. Problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$ has an alternative formulation as follows,

$$\begin{aligned} \mathcal{P}_{\text{cmmv}}^2(x_{\text{tg}}, q) : \quad & \min_{\{\mathbf{u}_t\}_{t=0}^{T-1}} \text{Var}_0[x_T] \\ & \text{s.t. } E_0[x_T] = x_{\text{tg}}, \\ & \{x_t, \mathbf{u}_t\} \text{ satisfy (3), (4).} \end{aligned} \quad (5)$$

where $x_{\text{tg}} > (\prod_{i=1}^T r_i)x_0$ is the target wealth level. Imposing the requirement $x_{\text{tg}} > \prod_{i=1}^T r_i$ is to avoid the degenerate case, i.e., the target should be at least greater than the investing all into a risk-free asset. Otherwise, the optimal policy is to invest all in risk-free assets. In the subsequent sections, we first develop the solution for problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$ and then extend the solution to the problem $\mathcal{P}_{\text{cmmv}}^2(x_{\text{tg}}, q)$.

3. Optimal portfolio policy

3.1. Portfolio policy for Markov regime-switching model

Since the problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$ is not separable in the context of dynamic programming, we adopt embedding methods (see Cui et al. (2022), Li and Ng (2000) and Pun and Ye (2022)) to solve it. Specifically, we first solve a parameterized auxiliary problem and then retrieve the solution to the original problem by identifying an appropriate parameter. We introduce the following auxiliary problem $\mathcal{A}^x(\omega, q, b)$ for problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$:

$$\begin{aligned} \mathcal{A}^x(\omega, q, b) : \quad & \min_{\{\mathbf{u}_t\}_{t=0}^{T-1}} E_0[(x_T - b)^2] - 2\omega E_0[x_T] \\ & \text{s.t. } \{x_t, \mathbf{u}_t\} \text{ satisfy (3), (4).} \end{aligned} \quad (6)$$

where $b \in \mathbb{R}$ is a parameter. Clearly, if $b = E_0[x_T]$, the term $E_0[(x_T - b)^2]$ becomes $\text{Var}_0[x_T]$ and the solution of problem $\mathcal{A}^x(\omega, q, b)$ solves problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$. The prominent feature of the auxiliary problem $\mathcal{A}^x(\omega, q, b)$ is that it is a linear-quadratic optimal control problem, which can be solved by dynamic programming.

⁴ Reader may refer to Bertsimas and Cory-Wright (2022) and Gao and Li (2013) and Bertsimas and Weismantel (2005) for the motivation of considering the cardinality constraint.

⁵ Notation $C_n^k = \frac{n(n-1)\dots(n-k+1)}{k(k-1)\dots 1}$ is the number of k -combinations.

We first rewrite problem $\mathcal{A}^x(\omega, q, b)$ in a more concise formulation. We define $\gamma_t \triangleq \prod_{k=t}^{T-1} r_{k+1}$ for $t = 0, \dots, T-1$ with $\gamma_T = 1$ as the cumulative risk-free return from t to T . We then change the state variable to

$$y_t \triangleq x_t - (b + \omega)/\gamma_t, \quad t = 0, \dots, T. \quad (7)$$

After changing the variable, the wealth dynamic (3) becomes $y_{t+1} = r_{t+1}(x_t - (b + \omega)/\gamma_t) + \mathbf{P}_{t+1}^\top \mathbf{u}_t = r_{t+1}y_t + \mathbf{P}_{t+1}^\top \mathbf{u}_t$, $t = 0, \dots, T-1$. Note that it has $y_T = x_T - (b + \omega)$. Adding the constant $2\omega b + \omega^2$ to the objective function (6) gives $E_0[(x_T - b)^2] - 2\omega E[x_T] + 2\omega b + \omega^2 = E[y_T^2]$, which does not change the problem $\mathcal{A}^x(\omega, q, b)$. Then problem $\mathcal{A}^x(\omega, q, b)$ can be rewritten as the following equivalent formulation by using state variable y_t ,

$$\begin{aligned} \mathcal{A}^y(\omega, q, b) \quad & \min_{\{\mathbf{u}_t\}_{t=0}^{T-1}} E_0[y_T^2] \\ \text{s.t.} \quad & y_{t+1} = r_{t+1}y_t + \mathbf{P}_{t+1}^\top \mathbf{u}_t, \quad t = 0, \dots, T-1, \\ & \mathbf{u}_t \in \mathbb{U}(q) \quad t = 0, \dots, T-1, \end{aligned}$$

where $y_0 = x_0 - (b + \omega)/\gamma_0$. Under the Markov regime-switching setting, i.e., Assumption 1, we denote the first and second moments of the excess return $\mathbf{P}_{t+1}$ as follows

$$\begin{aligned} \mathbf{d}_t(i) & \triangleq E_t[\mathbf{P}_{t+1} | s_t = i] \\ & = \boldsymbol{\mu}_t(i) - r_{t+1}\mathbf{e}, \quad i \in \mathbb{M}, \end{aligned} \quad (8)$$

$$\begin{aligned} \mathbf{D}_t(i) & \triangleq E_t[\mathbf{P}_{t+1} \mathbf{P}_{t+1}^\top | s_t = i] \\ & = \boldsymbol{\Sigma}_t(i) + \mathbf{d}_t(i)\mathbf{d}_t(i)^\top, \quad i \in \mathbb{M} \end{aligned} \quad (9)$$

for $t = 0, \dots, T-1$.

To solve problem $\mathcal{A}^y(\omega, q, b)$, we introduce a sequence of vectors, $\boldsymbol{\theta}_t = (\theta_t(1), \dots, \theta_t(m))^\top$ for $t = 0, \dots, T-1$. Specifically, the element $\theta_t(i)$ corresponds to market state $i \in \mathbb{M}$ and is recursively computed backward for $t = T-1, \dots, 0$ as follows:

$$\begin{aligned} \theta_t(i) & = \left(\sum_{j=1}^m p_t(i, j) \theta_{t+1}(j) \right) \left(1 - (\text{diag}(\mathbf{z}_t^*(i)) \mathbf{d}_t(i))^\top \right. \\ & \quad \times \left. (\text{diag}(\mathbf{z}_t^*(i)) \mathbf{D}_t(i) \text{diag}(\mathbf{z}_t^*(i)))^\dagger (\text{diag}(\mathbf{z}_t^*(i)) \mathbf{d}_t(i)) \right) \end{aligned} \quad (10)$$

for $i \in \mathbb{M}$, where $\theta_T(j) = 1$ for $j \in \mathbb{M}$ and $\mathbf{z}_t^*(i)$ is the optimal solution of the following problem:

$$\begin{aligned} \mathbf{z}_t^*(i) & = \arg \min_{\mathbf{z} \in \mathbb{B}(q)} g(\mathbf{z}) \triangleq -(\text{diag}(\mathbf{z}) \mathbf{d}_t(i))^\top \\ & \quad \times (\text{diag}(\mathbf{z}) \mathbf{D}_t(i) \text{diag}(\mathbf{z}))^\dagger (\text{diag}(\mathbf{z}) \mathbf{d}_t(i)). \end{aligned} \quad (11)$$

The procedure of computing $\boldsymbol{\theta}_t$ defines a mapping from $\boldsymbol{\theta}_{t+1}$ to $\boldsymbol{\theta}_t$ for all $t = T-1, \dots, 0$, involving a crucial step that solves problem (11) for each state $i \in \mathbb{M}$ at each time $t = T-1, \dots, 0$. Moreover, problem (11) constitutes a challenging nonlinear binary optimization problem. Using an enumeration method, where each cardinality index $\mathbf{z} \in \mathbb{B}(q)$ is enumerated, leads to evaluating $|\mathbb{B}(q)| = \sum_{k=1}^q C_n^k$ potential solutions, incurring significant computational costs. To expedite computation, we may leverage the following result to reframe problem (11) as a cardinality-constrained quadratic optimization (CCQO) problem.

Lemma 2. Given the parameters $\mathbf{H} \in \mathbb{S}_+^n$, $\mathbf{h} \in \mathbb{R}^n$ and $q \in \{1, \dots, n\}$, the following two problems:

$$\begin{aligned} \mathcal{Q}^{\text{ccqo}}(\mathbf{H}, \mathbf{h}, q) : \quad & \min_{\mathbf{x} \in \mathbb{R}^n} \mathbf{x}^\top \mathbf{H} \mathbf{x} + 2\mathbf{h}^\top \mathbf{x} \\ \text{s.t.} \quad & \|\mathbf{x}\|_0 \leq q, \end{aligned} \quad (12)$$

and

$$\begin{aligned} \mathcal{Q}^{\text{nbo}}(\mathbf{H}, \mathbf{h}, q) : \quad & \min_{\mathbf{z} \in \mathbb{B}(q)} -(\text{diag}(\mathbf{z}) \mathbf{h})^\top \\ & \times (\text{diag}(\mathbf{z}) \mathbf{H} \text{diag}(\mathbf{z}))^\dagger (\text{diag}(\mathbf{z}) \mathbf{h}) \end{aligned}$$

are equivalent in sense that the solutions of problem $\mathcal{Q}^{\text{ccqo}}(\mathbf{H}, \mathbf{h}, q)$ and problem $\mathcal{Q}^{\text{nbo}}(\mathbf{H}, \mathbf{h}, q)$ which are denoted by $\mathbf{x}^*$ and $\mathbf{z}^*$, respectively, satisfy the following relationship:

$$\begin{cases} \mathbf{x}^* = -(\text{diag}(\mathbf{z}^*) \mathbf{H} \text{diag}(\mathbf{z}^*))^\dagger (\text{diag}(\mathbf{z}^*) \mathbf{h}), \\ \mathbf{z}^* = \mathcal{I}(\mathbf{x}^*). \end{cases} \quad (13)$$

Furthermore, the objective values of the two problems are identical, i.e., $v(\mathcal{Q}^{\text{ccqo}}(\mathbf{H}, \mathbf{h}, q)) = v(\mathcal{Q}^{\text{nbo}}(\mathbf{H}, \mathbf{h}, q))$.

Proof. As the cardinality constraint $\|\mathbf{x}\|_0 \leq q$ limits the active elements of $\mathbf{x}$ in problem $\mathcal{Q}^{\text{ccqo}}(\mathbf{H}, \mathbf{h}, q)$, we may introduce additional binary variable $\mathbf{z} \in \mathbb{B}(q)$ and write the problem $\mathcal{Q}^{\text{ccqo}}(\mathbf{H}, \mathbf{h}, q)$ in the following equivalent form,

$$\begin{aligned} \mathcal{Q}^{\text{ccqo}}(\mathbf{H}, \mathbf{h}, q) \quad & \min_{\mathbf{x}, \mathbf{z}} f(\mathbf{z}, \mathbf{x}) = \mathbf{x}^\top (\text{diag}(\mathbf{z}) \mathbf{H} \text{diag}(\mathbf{z})) \mathbf{x} \\ & + 2(\text{diag}(\mathbf{z}) \mathbf{h})^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{z} = (z_1, \dots, z_n)^\top \in \mathbb{B}(q). \end{aligned} \quad (14)$$

In the above formulation, the term $\text{diag}(\mathbf{z}) \mathbf{H} \text{diag}(\mathbf{z})$ (or $\text{diag}(\mathbf{z}) \mathbf{h}$) yields a sparse matrix (or sparse vector) where the rows and columns indicated by $\mathbf{z}$ are selected, and all others are set to zero. For any fixed $\mathbf{z} \in \mathbb{B}(q)$, we first minimize $f(\mathbf{z}, \mathbf{x})$ with respect to $\mathbf{x}$. As the problem is convex optimization (for fixed $\mathbf{z}$), the first-order optimality condition is

$$\begin{aligned} \nabla_{\mathbf{x}} f(\mathbf{z}, \mathbf{x}) & = 2(\text{diag}(\mathbf{z}) \mathbf{H} \text{diag}(\mathbf{z})) \mathbf{x} + 2(\text{diag}(\mathbf{z}) \mathbf{h}) = 0 \\ \Rightarrow \mathbf{x}(\mathbf{z}) & = \arg \min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{z}, \mathbf{x}) \\ & = -(\text{diag}(\mathbf{z}) \mathbf{H} \text{diag}(\mathbf{z}))^\dagger (\text{diag}(\mathbf{z}) \mathbf{h}). \end{aligned} \quad (15)$$

Substituting $\mathbf{x}(\mathbf{z})$ back to (14) gives

$$\begin{aligned} f(\mathbf{z}, \mathbf{x}(\mathbf{z})) & = -(\text{diag}(\mathbf{z}) \mathbf{h})^\top \\ & \times (\text{diag}(\mathbf{z}) \mathbf{H} \text{diag}(\mathbf{z}))^\dagger (\text{diag}(\mathbf{z}) \mathbf{h}), \end{aligned} \quad (16)$$

which is the objective function of problem $\mathcal{Q}^{\text{nbo}}(\mathbf{H}, \mathbf{h}, q)$. If $\mathbf{z}^* = \arg \min_{\mathbf{z} \in \mathbb{B}(q)} f(\mathbf{z}, \mathbf{x}(\mathbf{z}))$ (i.e., solves problem $\mathcal{Q}^{\text{nbo}}(\mathbf{H}, \mathbf{h}, q)$) then Eq. (15) implies the result (13). $\square$

Lemma 2 basically says that the nonlinear binary optimization problem $\mathcal{Q}^{\text{nbo}}(\mathbf{H}, \mathbf{h}, q)$ has an equivalent CCQO formulation. Using such a result, we may obtain $\mathbf{z}_t^*$ in problem (11) by solving the following problem,

$$\begin{aligned} \mathcal{P}^{\text{ccqo}}(\mathbf{D}_t(i), \mathbf{d}_t(i), q) : \quad & \min_{\mathbf{k} \in \mathbb{R}^n} 2\mathbf{d}_t(i)^\top \mathbf{k} + \mathbf{k}^\top \mathbf{D}_t(i) \mathbf{k} \\ \text{s.t.} \quad & \|\mathbf{k}\|_0 \leq q, \end{aligned}$$

and then let $\mathbf{z}_t^*(i) = \mathcal{I}(\mathbf{k}_t^*(i))$ where $\mathbf{k}_t^*(i)$ is the optimal solution of problem $\mathcal{P}^{\text{ccqo}}(\mathbf{D}_t(i), \mathbf{d}_t(i), q)$.

Before we illustrate the main result, we provide some properties of variable $\{\boldsymbol{\theta}_t\}_{t=0}^{T-1}$.

Lemma 3.

- (i) Under Assumption 1 and condition (1), the elements of vector $\boldsymbol{\theta}_t = (\theta_t(1), \dots, \theta_t(m))^\top$ satisfy $0 < \theta_t(i) < 1$ for $i \in \mathbb{M}$ and $t = 0, \dots, T-1$.
- (ii) Given $q, \hat{q} \in \{1, \dots, n\}$ with $q < \hat{q}$, for any $t = 0, \dots, T-1$ and $i \in \mathbb{M}$, it has $\theta_t(i)|_q \geq \theta_t(i)|_{\hat{q}}$, where $\theta_t(i)|_q$ and $\theta_t(i)|_{\hat{q}}$ are the associated variables for q and $\hat{q}$ in $\mathbb{B}(\cdot)$ of $\mathbf{z}^*(i)$ defined in (10), respectively.

Proof. (i) We prove the result $0 < \theta_t(i) < 1$ for all $i \in \mathbb{M}$ and $t = T-1, T-2, \dots, 0$ by using mathematical induction method. Before we proceed to the proof, we first introduce the following notation. For a symmetrical matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ (or for a vector $\mathbf{a} \in \mathbb{R}^n$), given some $\mathbf{z} = (z_1, \dots, z_n)^\top \in \mathbb{B}(q)$, the notations $\mathbf{A}[\mathbf{z}]$ (or $\mathbf{a}[\mathbf{z}]$) denotes the submatrix (or vector) constructed by the rows and columns indicated by the $z_i = 1$ for $i = 1, \dots, n$ in vector $\mathbf{z}$.⁶ The Assumption 1 implies that $\Sigma_t(i) = \mathbf{D}_t(i) - \mathbf{d}_t(i)\mathbf{d}_t(i)^\top > \mathbf{0}$ for all $t = 0, \dots, T-1$ and $i \in \mathbb{M}$, which implies that its submatrix satisfies (see Horn and Johnson (2012)):

$$\Sigma_t(i)[\mathbf{z}] = \mathbf{D}_t(i)[\mathbf{z}] - \mathbf{d}_t(i)[\mathbf{z}]\mathbf{d}_t(i)[\mathbf{z}]^\top > \mathbf{0} \quad (17)$$

for any $\mathbf{z} \in \mathbb{B}(q)$, $i \in \mathbb{M}$ and $t = 0, \dots, T-1$. Using (17) and the Schur's complement theorem (Zhang, 2006), it has

$$\begin{pmatrix} \mathbf{D}_t(i)[\mathbf{z}] & \mathbf{d}_t(i)[\mathbf{z}] \\ \mathbf{d}_t(i)[\mathbf{z}]^\top & 1 \end{pmatrix} > \mathbf{0} \\ \Leftrightarrow 1 - \mathbf{d}_t(i)[\mathbf{z}]^\top \mathbf{D}_t(i)[\mathbf{z}]^{-1} \mathbf{d}_t(i)[\mathbf{z}] > 0, \quad (18)$$

for any $\mathbf{z} \in \mathbb{B}(q)$, $i \in \mathbb{M}$ and $t = 0, \dots, T-1$, which further implies

$$\begin{aligned} 1 &> 1 - (\text{diag}(\mathbf{z})\mathbf{d}_t(i))^\top (\text{diag}(\mathbf{z})\mathbf{D}_t(i)\text{diag}(\mathbf{z}))^\dagger \\ &\quad \times (\text{diag}(\mathbf{z})\mathbf{d}_t(i)) \\ &= 1 - \mathbf{d}_t(i)[\mathbf{z}]^\top \mathbf{D}_t(i)[\mathbf{z}]^{-1} \mathbf{d}_t(i)[\mathbf{z}] > 0, \end{aligned} \quad (19)$$

for any $\mathbf{z} \in \mathbb{B}(q)$, $i \in \mathbb{M}$ and $t = 0, \dots, T-1$. In (19), the equality is from the fact that eliminating all the zero elements in the matrices and vector does not change the value of their product. Furthermore, the first inequality is due to the fact $\mathbf{D}_t(i)[\mathbf{z}] > \mathbf{0}$ for all $\mathbf{z} \in \mathbb{B}(q)$ and the last inequality is due to the inequality (18). At $t = T-1$, since $\theta_T(j) = 1$ for $j \in \mathbb{M}$, the definition of $\theta_{T-1}(i)$ and inequality (19) implies that $0 < \theta_{T-1}(i) < 1$ for all $i \in \mathbb{M}$. We then assume $0 < \theta_{t+1}(i) < 1$ for all $i \in \mathbb{M}$. At time t , it has $0 < \sum_{j=1}^m p_{ij}\theta_{t+1}(j) < 1$ for all $i \in \mathbb{M}$. The definition (10) and inequality (19) imply that $0 < \theta_t(i) < 1$ for all $i \in \mathbb{M}$, completes our proof of result (i).

We then focus on result (ii). Using the mathematical induction method, we first prove the result at time $t = T-1$. For problem (11), we use $\mathbf{z}_{T-1}^*(i)|_q$ and $\mathbf{z}_{T-1}^*(i)|_{\hat{q}}$ to denote the optimizers to problem (11) for parameters being q and $\hat{q}$, respectively. Since the feasible sets satisfy $\mathbb{B}(q) \subset \mathbb{B}(\hat{q})$, the objective values satisfy $g(\mathbf{z}_{T-1}^*(i)|_{\hat{q}}) \leq g(\mathbf{z}_{T-1}^*(i)|_q)$, which further implies $\theta_{T-1}(i)|_{\hat{q}} \leq \theta_{T-1}(i)|_q$ for all $i \in \mathbb{M}$. Then we assume at $t+1$, it has $\theta_{t+1}(i)|_{\hat{q}} \leq \theta_{t+1}(i)|_q$ for all $i \in \mathbb{M}$. At time t , it has $0 < \sum_{j=1}^m p_{t,i}(j)\theta_{t+1}(j)|_{\hat{q}} \leq \sum_{j=1}^m p_{t,i}(j) \cdot \theta_{t+1}(j)|_q$. Furthermore, we may apply the similar argument at $t = T-1$, which gives $g(\mathbf{z}_t^*(j)|_{\hat{q}}) \leq g(\mathbf{z}_t^*(j)|_q)$ for all $j \in \mathbb{M}$. Then the Eq. (10) implies that $\theta_t(i)|_{\hat{q}} \leq \theta_t(i)|_q$ for all $i \in \mathbb{M}$. $\square$

With the help of the parameters $\{\theta_t\}_{t=0}^T$, the optimal solution of problem $\mathcal{A}^y(\omega, q, b)$ is characterized by the following result.

Proposition 4. The optimal portfolio policy of problem $\mathcal{A}^y(\omega, q, b)$ is

$$\mathbf{u}_t^*(i) = r_{t+1}y_t \mathbf{k}_t^*(i), \quad t = 0, \dots, T-1, \quad (20)$$

where $\mathbf{k}_t^*(i)$ is the optimal solution of problem $\mathcal{P}^{\text{ccqo}}(\mathbf{D}_t(i), \mathbf{d}_t(i), q)$ when market state is $s_t = i$ for $i \in \mathbb{M}$ for $t = 0, \dots, T-1$. Moreover, the optimal objective value of problem $\mathcal{A}^y(\omega, q, b)$ is

$$v(\mathcal{A}^y(\omega, q, b)) = \theta_0(s_0)\gamma_0^2 y_0^2. \quad (21)$$

⁶ We use one example to illustrate this notation. Given matrix $\mathbf{A} = \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{pmatrix}$ and $\mathbf{z} = (1, 0, 1)$, then $\mathbf{A}[\mathbf{z}] = \begin{pmatrix} a_{1,1} & & a_{1,3} \\ & & a_{3,1} \\ a_{3,1} & & a_{3,3} \end{pmatrix}$.

Proof. We solve problem $\mathcal{A}^y(\omega, q, b)$ by dynamic programming. For any time $t \in \{0, 1, 2, \dots, T-1\}$, given $s_t \in \mathbb{M}$, we introduce the truncated version of problem $\mathcal{A}^y(\omega, q, b)$ as follows,

$$\mathcal{A}^y(\omega, q, b)|_t^T : V_t(y_t, s_t) = \min_{\{\mathbf{u}_k\}_{k=t}^{T-1}} \mathbb{E}_t[y_T^2] \quad (22)$$

$$\begin{aligned} \text{s.t. } y_{k+1} &= r_{k+1}y_k + \mathbf{P}_{k+1}^\top \mathbf{u}_k, \quad k = t, t+1, \dots, T-1, \\ \mathbf{u}_k &\in \mathbb{U}(q), \quad k = t, t+1, \dots, T-1. \end{aligned}$$

Problem $\mathcal{A}^y(\omega, q, b)|_t^T$ shares a similar structure as the original problem $\mathcal{A}^y(\omega, q, b)$, albeit with a shorter horizon starting at time t and the initial state being y_t . The optimal value of problem $\mathcal{A}^y(\omega, q, b)|_t^T$ defines the value function for the dynamic programming algorithm (Bertsekas, 2005), which satisfies the following recursive equation:

$$\begin{aligned} V_t(y_t, s_t) &= \min_{\mathbf{u}_t \in \mathbb{U}(q)} \mathbb{E}_t[V_{t+1}(y_{t+1}, s_{t+1})] \\ &= \min_{\mathbf{u}_t \in \mathbb{U}(q)} \mathbb{E}_t[V_{t+1}(r_{t+1}y_t + \mathbf{P}_{t+1}^\top \mathbf{u}_t, s_{t+1})]. \end{aligned} \quad (23)$$

We claim that the value function (22) can be written in the following form,

$$V_t(y_t, s_t) = \theta_t(s_t)\gamma_t^2 y_t^2, \quad t = 0, \dots, T, \quad (24)$$

where $\theta_t(s_t) \in \mathbb{R}$ is defined in (10).

We adopt the mathematical induction method to prove the claim (24). At time $t = T$, the claim (24) is automatically true as $\theta_T(j) = 1$ for $j \in \mathbb{M}$. Then, we suppose the claim (24) is true at time $t+1$ and prove the claim (24) also holds at time t . Under assumption (24) at time $t+1$, if $s_t = i$, Eq. (23) becomes

$$\begin{aligned} V_t(y_t, i) &= \min_{\mathbf{u}_t \in \mathbb{U}(q)} \mathbb{E}_t \left[\theta_{t+1}(s_{t+1}) \gamma_{t+1}^2 (r_{t+1}y_t + \mathbf{P}_{t+1}^\top \mathbf{u}_t)^2 \right] \\ &= \min_{\mathbf{u}_t \in \mathbb{U}(q)} \gamma_{t+1}^2 \mathbb{E}_t [\theta_{t+1}(s_{t+1})] \left(r_{t+1}^2 y_t^2 \right. \\ &\quad \left. + 2r_{t+1}y_t \mathbb{E}_t [\mathbf{P}_{t+1}^\top \mathbf{u}_t] + \mathbf{u}_t^\top \mathbb{E}_t [\mathbf{P}_{t+1} \mathbf{P}_{t+1}^\top] \mathbf{u}_t \right) \\ &= \gamma_{t+1}^2 c_t(i) \left(r_{t+1}^2 y_t^2 + \min_{\mathbf{u}_t \in \mathbb{U}(q)} \left\{ 2r_{t+1}y_t \mathbf{d}_t(i)^\top \mathbf{u}_t \right. \right. \\ &\quad \left. \left. + \mathbf{u}_t^\top \mathbf{D}_t(i) \mathbf{u}_t \right\} \right), \end{aligned} \quad (25)$$

where $c_t(i) = \sum_{j=1}^m p_{t,i}(j)\theta_{t+1}(j)$, $\mathbf{d}_t(i)$ and $\mathbf{D}_t(i)$ are defined in (8) and (9), respectively. As the state variable y_t is a random variable at time $t = 0$, we cannot solve this problem offline. Fortunately, we may apply Lemma 2 to this problem, i.e., the optimization problem in (25) can be regarded as problem $\mathcal{Q}^{\text{ccqo}}(\mathbf{D}_t(i), r_{t+1}y_t \mathbf{d}_t(i), q)$. Then, the Eq. (13) implies that the optimization problem in (25) is

$$\begin{aligned} \mathbf{u}_t(i; \mathbf{z}^*) &= -r_{t+1}y_t (\text{diag}(\mathbf{z}^*)\mathbf{D}_t(i)\text{diag}(\mathbf{z}^*))^\dagger \\ &\quad \times (\text{diag}(\mathbf{z}^*)\mathbf{d}_t(i)), \end{aligned} \quad (26)$$

where $\mathbf{z}^* \in \mathbb{B}(q)$ is some binary vector. That is to say, the optimal solution of the problem (25) is in a linear function of y_t . Motivated by such a structure, we regard $\mathbf{k}_t$ as an unknown vector and assume the portfolio policy is $\mathbf{u}_t = r_{t+1}y_t \mathbf{k}_t$. Substituting $\mathbf{u}_t$ in problem (25) gives

$$\begin{aligned} V_t(y_t, i) &= \gamma_{t+1}^2 r_{t+1}^2 y_t^2 c_t(i) \left(1 + \min_{\mathbf{k} \in \mathbb{U}(q)} \left\{ 2\mathbf{d}_t(i)^\top \mathbf{k} \right. \right. \\ &\quad \left. \left. + \mathbf{k}^\top \mathbf{D}_t(i) \mathbf{k} \right\} \right), \end{aligned} \quad (27)$$

If we ignore the constant in (27), it gives the problem $\mathcal{P}^{\text{ccqo}}(\mathbf{D}_t(i), \mathbf{d}_t(i), q)$ which further determines the optimal solution $\mathbf{k}_t^*(i)$. Since problem $\mathcal{P}^{\text{ccqo}}(\mathbf{D}_t(i), \mathbf{d}_t(i), q)$ is independent to y_t , it can solved offline. Then, by using Lemma 2 again, the value

function (27) can be written as

$$V_t(y_t, i) = \gamma_{t+1}^2 r_{t+1}^2 y_t^2 c_t(i) \left(1 - (\text{diag}(\mathbf{z}^*) \mathbf{d}_t(i))^\top \right. \\ \left. \times (\text{diag}(\mathbf{z}^*) \mathbf{D}_t(i) \text{diag}(\mathbf{z}^*))^\dagger (\text{diag}(\mathbf{z}^*) \mathbf{d}_t(i)) \right) = \gamma_t^2 y_t^2 c_t(i)$$

where $\mathbf{z}^* = \mathcal{I}(\mathbf{k}_t^*(i))$ and $\mathbf{k}_t^*(i)$ is the solution of optimization problem in (27). Thus, we obtain $\theta_t(i)$ in (10) which proves the claim (24). Using $\mathbf{k}_t^*(i)$, we get optimal portfolio $\mathbf{u}_t^*(i) = r_{t+1} y_t \mathbf{k}_t^*(i)$ as in (20). $\square$

With the help of solution of problem $\mathcal{A}^y(\omega, q, b)$, we have the following result for problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$.

Theorem 5. Under Assumption 1, when the market state is $s_t = i$ at time t , the optimal portfolio policy of problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$ is

$$\mathbf{u}_t^*(i) = r_{t+1} \left(x_t - \frac{\gamma_0 x_0 \theta_0(s_0) + \omega}{\theta_0(s_0) \gamma_t} \right) \mathbf{k}_t^*(i), \quad (28)$$

for $t = 0, \dots, T-1$, where $\mathbf{k}_t^*(i)$ is solution of problem $\mathcal{P}^{\text{ccqo}}(\mathbf{D}_t(i), \mathbf{d}_t(i), q)$ and $\theta_0(s_0)$ is defined in (10). Implementing the optimal policy (28), the expected value and variance of the terminal wealth are

$$E_0[x_T^*] = \gamma_0 x_0 + \omega \left(\frac{1}{\theta_0(s_0)} - 1 \right) \\ \text{Var}_0[x_T^*] = \omega^2 \left(\frac{1}{\theta_0(s_0)} - 1 \right), \quad (29)$$

respectively; and the MV efficient frontier can be expressed as

$$\text{Var}_0[x_T^*] = \frac{\theta_0(s_0)(E_0[x_T^*] - \gamma_0 x_0)^2}{1 - \theta_0(s_0)}. \quad (30)$$

Proof. Using the expression (20) in Proposition 4 and the relationship (7), we know the optimal portfolio policy of problem $\mathcal{A}^x(\omega, q, b)$ is

$$\mathbf{u}_t^*(i) = r_{t+1} \left(x_t - \frac{b + \omega}{\gamma_t} \right) \mathbf{k}_t^*(i), \quad t = 0, \dots, T-1, \quad (31)$$

when the market state is $s_t = i$ where $\mathbf{k}_t^*(i)$ is the solution of problem $\mathcal{P}^{\text{ccqo}}(\mathbf{D}_t(i), \mathbf{d}_t(i), q)$.

On the other hand, checking the objective functions of problem $\mathcal{A}^x(\omega, q, b)$ and problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$, we know that if the parameter $b = b^*$ satisfies $b^* = E_0[x_T^*]$, where x_T^* is the terminal wealth resulted from using the optimal policy (31), then the objective functions of these two problems are equivalent. Thus, when $b^* = E_0[x_T^*]$, the policy of problem $\mathcal{A}^x(\omega, b)$ also solves problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$. Note that the expected value $E_0[x_T^*]$ is the minimizer of the problem $\min_b \{E_0[(x_T^* - b)^2] - 2\omega E_0[x_T^*]\}$. Thus, we may characterize b^* through the following optimization problem,

$$E_0[x_T^*] = \arg \min_b v(\mathcal{A}^x(\omega, q, b)) \\ = \arg \min_b \{E_0[(x_T^* - b)^2] - 2\omega E_0[x_T^*]\}. \quad (32)$$

That is to say, the minimizer of the above problem is the desired parameter b that satisfies $b = E_0[x_T^*]$. We then compute the value $v(\mathcal{A}^x(\lambda, q, b))$. Notably, the difference between the objective values of problem $\mathcal{A}^x(\omega, q, b)$ and $\mathcal{A}^y(\omega, q, b)$ is $2\omega b + \omega^2$. Using Eq. (7) and Proposition 4, it has

$$v(\mathcal{A}^x(\omega, q, b)) = v(\mathcal{A}^y(\omega, q, b)) - 2\omega b - \omega^2 \\ = \theta_0(s_0) \gamma_0^2 \left(x_0 - \frac{b + \omega}{\gamma_0} \right)^2 - 2\omega b - \omega^2, \quad (33)$$

which further implies that

$$E_0[x_T^*] = b^* = \arg \min_b \left\{ \theta_0(s_0) \gamma_0^2 \left(x_0 - \frac{b + \omega}{\gamma_0} \right)^2 \right. \\ \left. - 2\omega b - \omega^2 \right\} = \gamma_0 x_0 + \omega \left(\frac{1}{\theta_0(s_0)} - 1 \right). \quad (34)$$

Then we let $b = b^*$ in policy (31) which yields the optimal policy (28) for problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$.

We then derive the MV efficient frontier of problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$. The expected value of optimal terminal wealth can be computed by (32) and (34), which gives $E_0[x_T^*] = b^*$. We only need to compute the variance of x_T^* . As $\text{Var}_0[x_T] = v(\mathcal{A}^x(\omega, q, b)) + 2\omega E_0[x_T^*]$, using (33) and (34), it has

$$\text{Var}_0[x_T] = \omega^2 \left(\frac{1}{\theta_0(s_0)} - 1 \right),$$

which further proves the result (29). Eliminating ω in $E_0[x_T^*]$ and $\text{Var}_0[x_T^*]$ gives result (30). $\square$

We then extend the solution of problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$ to problem $\mathcal{P}_{\text{cmmv}}^2(x_{\text{tg}}, q)$.

Corollary 6. Under Assumption 1, given the market state as $s_t = i$ at time t , the optimal portfolio policy of problem $\mathcal{P}_{\text{cmmv}}^2(x_{\text{tg}}, q)$ is as same as Eq. (28) where the parameter ω is set as

$$\omega = \frac{\theta_0(s_0)(x_{\text{tg}} - \gamma_0 x_0)}{1 - \theta_0(s_0)}, \quad (35)$$

and $\mathbf{k}_t^*(i)$ is solution of problem $\mathcal{P}^{\text{ccqo}}(\mathbf{D}_t(i), \mathbf{d}_t(i), q)$ and $\theta_0(s_0)$ is defined in (10). Furthermore, the variance of the terminal wealth (the optimal objective value) in problem $(\mathcal{P}_{\text{cmmv}}^2(x_{\text{tg}}, q))$ is

$$\text{Var}_0[x_T^*] = \frac{\theta_0(s_0)(x_{\text{tg}} - \gamma_0 x_0)^2}{1 - \theta_0(s_0)}. \quad (36)$$

Proof. To solve problem $\mathcal{P}_{\text{cmmv}}^2(x_{\text{tg}}, q)$, we introduce Lagrange multiplier $\omega \geq 0$ for constraint (5) and consider the following problem,

$$\mathcal{L}(\omega) : \min_{\{\mathbf{u}_t\}_{t=0}^{T-1}} \text{Var}_0[x_T] - 2\omega(E_0[x_T] - x_{\text{tg}}) \quad (37) \\ (\text{s.t.}) \quad \{\mathbf{x}_t, \mathbf{u}_t\} \text{ satisfy (3), (4).}$$

Since any solution of problem $\mathcal{P}_{\text{cmmv}}^2(x_{\text{tg}}, q)$ is a feasible solution to problem $\mathcal{L}(\omega)$, the weak duality relationship implies $v(\mathcal{L}(\omega)) \leq v(\mathcal{P}_{\text{cmmv}}^2(x_{\text{tg}}, q))$. The strong duality holds when the expected wealth $E_0[x_T^*]$ generated from problem $\mathcal{L}(\omega)$ satisfies the constraint (5), i.e., $E_0[x_T^*] = x_{\text{tg}}$. Note that problem $\mathcal{L}(\omega)$ is similar to problem $\mathcal{P}_{\text{cmmv}}^1(\omega)$ (with the difference term $2\omega x_{\text{tg}}$). Thus, using (34), it has

$$E[x_T^*] = \gamma_0 x_0 + \omega \left(\frac{1}{\theta_0(s_0)} - 1 \right) = x_{\text{tg}}. \quad (38)$$

Solving the above equation for ω leads to the result (35). As problem $\mathcal{L}(\omega)$ and $\mathcal{P}_{\text{cmmv}}^1(\omega)$ have the identical optimal portfolio policy, substituting ω by Eq. (35) in policy (28) gives the optimal portfolio policy for problem $\mathcal{P}_{\text{cmmv}}^2(x_{\text{tg}}, q)$. The variance $\text{Var}_0[x_T^*]$ can be obtained in the similar way. $\square$

In Theorem 5 and Corollary 6, the key step for characterizing the optimal portfolio policy (28) involves solving problem $\mathcal{P}^{\text{ccqo}}(\mathbf{D}_t(i), \mathbf{d}_t(i), q)$, which is a cardinality-constrained quadratic optimization (CCQO) problem. While there are specifically designed algorithms for solving CCQO problems (see, e.g., Gao and Li (2013) and Bertsimas and Cory-Wright (2022)), one standard

method is to adopt the associated *mixed-integer quadratic programming* (MIQP) formulation given as follows:

$$\begin{aligned} \mathcal{P}_{\text{miqp}}^{\text{ccqo}}(\mathbf{D}_t(i), \mathbf{d}_t(i), q) : \\ \min_{\mathbf{k} \in \mathbb{R}^n, \mathbf{z} \in \{0,1\}^n} 2\mathbf{d}_t(i)^\top \mathbf{k} + \mathbf{k}^\top \mathbf{D}_t(i) \mathbf{k} \\ \text{s.t. } -z_l M \leq k_l \leq z_l M, \quad l = 1, \dots, n, \\ z_l \in \{0, 1\}, \quad l = 1, \dots, n, \\ \sum_{l=1}^n z_l \leq q, \end{aligned}$$

where $\mathbf{k} = (k_1, \dots, k_n)^\top$ and $M > 0$ is a large positive number. Problem $\mathcal{P}_{\text{miqp}}^{\text{ccqo}}(\mathbf{D}_t(i), \mathbf{d}_t(i), q)$ can be readily solved by commercial solvers such as [Gurobi Optimization \(2023\)](#). For typical portfolio sizes ($n \leq 100$) and practical cardinality constraints ($q \leq 20$), these MIQP solvers can efficiently resolve each sub-problem within reasonable time (just seconds under settings similar to our experiments), making our method computationally feasible for portfolio rebalancing at monthly or quarterly intervals. Furthermore, the CCQO sub-problems for different market states at the same time period are independent and can be solved in parallel, which further enhances computational efficiency.

We then provide some financial interpretation of the results given in [Theorem 5](#) and [Corollary 6](#). The CMMV portfolio (28) means that the proportion invested in each risky asset is totally determined by vector $\mathbf{k}_t^*(i)$, which is related to cardinality size q through the CCQO problem $\mathcal{P}_{\text{miqp}}^{\text{ccqo}}(\mathbf{D}_t(i), \mathbf{d}_t(i), q)$. Besides $\mathbf{k}_t^*(i)$, parameter q also affects $\theta_0(s_0)$, as shown in (10). This means that $\theta_0(s_0)$ controls the amount invested in risky assets, as the policy (28) indicates that the total amount invested in risky assets at time t is $\frac{1}{\gamma_t + 1}(\gamma_t x_t - \gamma_0 x_0 - \omega / \theta_0(s_0)) \mathbf{e}^\top \mathbf{k}_t^*(i)$. Besides the portfolio, the impact of q on the Sharpe ratio of the terminal wealth is given in the following result.

Corollary 7. Given $s_t \in \mathbb{M}$ and some $q \in \{1, \dots, n\}$ at time $t = 0, \dots, T-1$, we define the conditional Sharpe ratio generated from problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$ as $SR_t(s_t, q) \triangleq (E_t[x_T | s_t] - \gamma_t x_t) / \sqrt{\text{Var}_t[x_T | s_t]}$. Then, we have

$$SR_t(s_t, q) = \sqrt{\frac{1}{\theta_t(s_t)} - 1}. \quad (39)$$

Furthermore, if $\hat{q}, q \in \{1, \dots, n\}$ and $\hat{q} > q$, we have $SR_t(s_t, \hat{q}) > SR_t(s_t, q)$.

The result (10) is well-defined since $\theta_t(s_t) \in (0, 1)$ for $t = 0, \dots, T-1$ due to [Lemma 3](#). The proof of [Corollary 7](#) is straightforward. We consider a truncated version of problem $\mathcal{P}_{\text{cmmv}}(\omega, q)$ which starts from time t with given wealth x_t and market condition $s_t = i$. Applying the similar proof of [Theorem 5](#) to this truncated problem, we may obtain the expression of the MV efficient frontier similar to (30) as $\text{Var}_t[x_T^* | s_t] = (\theta_t(s_t)(E_t[x_T^* | s_t] - \gamma_t x_t))^2 / (1 - \theta_t(s_t))$, which further leads to the result (39). Thus, combining result (ii) in [Lemma 3](#) and Eq. (39), it has the monotonicity result $SR_t(s_t, \hat{q}) > SR_t(s_t, q)$ if $\hat{q} > q$ for given state s_t and x_t . Although Corollary implies larger cardinality size q leads to a better Sharpe ratio, there is a tradeoff as larger q also leads to more active assets to handle, which may lead to higher transaction costs and management costs (see [Table 1](#) in [Section 4](#)).

Table 1

Transaction cost (TC) and Sharpe ratio (SR) of varying cardinality q .

q	CMV-static		CMMV-independent		CMMV	
	TC	SR	TC (10^{-4})	SR	TC (10^{-4})	SR
1	0	0.152	2.272	0.157	10.828	0.157
2	0	0.189	3.270	0.190	15.985	0.190
3	0	0.205	3.945	0.192	4.280	0.214
4	0	0.209	4.680	0.207	4.936	0.228
5	0	0.221	5.005	0.217	9.050	0.231
6	0	0.228	5.328	0.219	5.915	0.234
7	0	0.228	5.434	0.216	6.203	0.231
8	0	0.228	5.815	0.212	6.695	0.228
9	0	0.228	6.185	0.209	6.510	0.223
10	0	0.228	6.228	0.205	6.590	0.220

3.2. Special settings: market with time-invariant parameters and the independent returns

In the Markov regime-switching market (i.e., [Assumption 1](#)), if we further assume that the market parameters are time-invariant, i.e., the random returns of the assets are stationary, then the solution process for $\{\theta_t\}_{t=0}^{T-1}$ can be simplified. Specifically, we consider the following assumption.

Assumption 8. Under [Assumption 1](#), we further assume all market parameters are time-invariant. That is, there exist $\mathbf{d}(i) \in \mathbb{R}^n$ and $\mathbf{D}(i) \in \mathbb{S}_+^n$ for $i \in \mathbb{M}$, and $p(i, j)$ for $i, j \in \mathbb{M}$, such that $\mathbf{d}_t(i) = \mathbf{d}(i)$, $\mathbf{D}_t(i) = \mathbf{D}(i)$ for $i \in \mathbb{M}$, and $p_t(i, j) = p(i, j)$ for $i, j \in \mathbb{M}$, for $t = 0, \dots, T-1$.

Under [Assumption 8](#), the optimization problem in (11) (or equivalently, problem $\mathcal{P}_t^{\text{ccqo}}(\mathbf{D}(i), \mathbf{d}(i), q)$) is independent of the time index t . Thus, we can solve the following problem for each state $i \in \mathbb{M}$:

$$\begin{aligned} \mathcal{P}_{t\text{-inv}}^{\text{ccqo}}(\mathbf{D}(i), \mathbf{d}(i), q) : \min_{\mathbf{k} \in \mathbb{R}^n} 2\mathbf{d}(i)^\top \mathbf{k} + \mathbf{k}^\top \mathbf{D}(i) \mathbf{k} \\ \text{s.t. } \|\mathbf{k}\|_0 \leq q, \end{aligned}$$

where the optimal solution is $\mathbf{k}^*(i)$ with the associated optimal cardinality set $\mathbf{z}^*(i) = \mathcal{I}(\mathbf{k}^*(i))$. We then define the vector $\boldsymbol{\rho} = (\rho(1), \dots, \rho(m))^\top$ with the element being

$$\begin{aligned} \rho(i) = 1 - (\text{diag}(\mathbf{z}^*(i))\mathbf{d}(i))^\top (\text{diag}(\mathbf{z}^*(i))\mathbf{D}(i)\text{diag}(\mathbf{z}^*(i)))^\dagger \\ \times (\text{diag}(\mathbf{z}^*(i))\mathbf{d}(i)), \quad i \in \mathbb{M} \end{aligned}$$

The parameter θ_t defined in (10) can be simplified as

$$\theta_t(i) = \sum_{j=1}^m p(i, j) \theta_{t+1}(j) \rho(j), \quad i \in \mathbb{M}, \quad t = T-1, \dots, 0, \quad (40)$$

with $\theta_T(i) = 1$ for $i \in \mathbb{M}$. Using $\mathbf{Q} = \{p(i, j)\}_{i, j \in \mathbb{M}}$ to denote the probability transition matrix of the Markov chain, we rewrite (40) by the vector formulation:

$$\begin{aligned} \boldsymbol{\theta}_t = \text{diag}(\boldsymbol{\rho})\mathbf{Q}\boldsymbol{\theta}_{t+1}, \quad t = T-1, \dots, 0, \\ \Rightarrow \boldsymbol{\theta}_0 = (\text{diag}(\boldsymbol{\rho})\mathbf{Q})^T \mathbf{e}. \end{aligned} \quad (41)$$

The above result indicates that we only need to solve problem $\mathcal{P}_{t\text{-inv}}^{\text{ccqo}}(\mathbf{D}(i), \mathbf{d}(i), q)$ for $i \in \mathbb{M}$ to obtain $\boldsymbol{\rho}$ and then compute $\boldsymbol{\theta}_0$ directly from (41).

Then the portfolio policy of problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$ can be simplified as follows.

Corollary 9. Under [Assumption 8](#), when the market state is $s_t = i$ at time t , the optimal portfolio policy of problem $\mathcal{P}_{\text{cmmv}}^1(\omega, q)$ is

$$\mathbf{u}_t^*(i) = r_{t+1} \left(x_t - \frac{\gamma_0 x_0 \theta_0(s_0) + \omega}{\theta_0(s_0) \gamma_t} \right) \mathbf{k}^*(i), \quad (42)$$

for $t = 0, \dots, T-1$, where $\mathbf{k}^*(i)$ is solution of problem $\mathcal{P}_{t-\text{inv}}^{\text{ccqo}}(\mathbf{D}(i), \mathbf{d}(i), q)$ and $\theta_0(s_0)$ is defined in (41). Implementing the optimal policy (41), the expected value and variance of the terminal wealth are in the similar formulation as Eq. (29). The MV efficient frontier can be expressed as Eq. (30).

The time invariant optimal portfolio policy of problem $\mathcal{P}_{\text{cmmv}}^2(x_{\text{tg}})$ is similar to the one in Corollary 6 with $\theta_0(s_0)$ be replaced by (41).

Previous studies have developed the optimal portfolio policy for the CMMV problem when the assets' returns follow the Markov regime-switching model. We then examine a special market setting that assumes assets have independent returns over time. Such an assumption has been widely adopted in dynamic portfolio optimization (see, e.g., Cui et al. (2014), Li and Ng (2000) and Cui et al. (2022)), which focuses on cases where the correlation of returns over time is weak.

Assumption 10. The expected value and covariance matrix of the random return $\mathbf{R}_t$ are independent over time for $t = 1, \dots, T$.

Under Assumption 10, the expected value and the covariance matrix of the excess return $\mathbf{P}_t$ are $\hat{\mathbf{d}}_t \triangleq \mathbb{E}[\mathbf{P}_{t+1}] = \boldsymbol{\mu}_t - r_{t+1}\mathbf{e}$ and $\hat{\mathbf{D}}_t \triangleq \mathbb{E}[\mathbf{P}_{t+1}\mathbf{P}_{t+1}^\top] = \boldsymbol{\Sigma}_t + \mathbf{d}_t\mathbf{d}_t^\top$, respectively, where $\boldsymbol{\mu}_t = \mathbb{E}[\mathbf{R}_{t+1}]$ and $\boldsymbol{\Sigma}_t = \text{Cov}[\mathbf{R}_{t+1}] = \mathbb{E}[\mathbf{R}_{t+1}\mathbf{R}_{t+1}^\top] - \boldsymbol{\mu}_t\boldsymbol{\mu}_t^\top$ for $t = 1, \dots, T$.

As the assumption of independent returns can be viewed as a special case of the Markov regime-switching returns, we may modify the solution approach developed in Section 3.1 to solve problem $\mathcal{P}_{\text{cmmv}}^1(\omega)$. Under Assumption 10, we introduce the following parameter $\hat{\theta}_t \in \mathbb{R}$ for $t = 0, \dots, T$ as follows,

$$\hat{\theta}_t = \hat{\theta}_{t+1} \cdot \left(1 - (\text{diag}(\hat{\mathbf{z}}_t^*)\hat{\mathbf{d}}_t)^\top (\text{diag}(\hat{\mathbf{z}}_t^*)\hat{\mathbf{D}}_t \text{diag}(\hat{\mathbf{z}}_t^*))^\dagger \times (\text{diag}(\hat{\mathbf{z}}_t^*)\hat{\mathbf{d}}_t) \right), \quad t = T-1, \dots, 0, \quad (43)$$

where $\hat{\theta}_T = 1$, $\hat{\mathbf{z}}_t^* = \mathcal{I}(\hat{\mathbf{k}}_t^*)$ is optimal cardinality and $\hat{\mathbf{k}}_t^*$ is the optimal solution of the problem $\mathcal{P}_{\text{miqp}}^{\text{ccqo}}(\hat{\mathbf{D}}_t, \hat{\mathbf{d}}_t, q)$. Then the following result characterizes the optimal portfolio policy of problem $\mathcal{P}_{\text{cmmv}}^1(\omega)$ under the independent assumption.

Theorem 11. Under Assumption 10, the optimal portfolio policy for problem $(\mathcal{P}_{\text{cmmv}}^1)$ is

$$\hat{\mathbf{u}}_t^* = r_{t+1} \left(x_t - \frac{\gamma_0 x_0 \hat{\theta}_0 + \omega}{\hat{\theta}_0 \gamma_t} \right) \hat{\mathbf{k}}_t^*, \quad (44)$$

for $t = 0, \dots, T-1$, where $\hat{\mathbf{k}}_t^*$ is solution of problem $\hat{\mathcal{P}}_{\text{miqp}}^{\text{ccqo}}(\hat{\mathbf{D}}_t, \hat{\mathbf{d}}_t, q)$ and $\hat{\theta}_0$ is defined in (43). Implementing the optimal policy (44), the expected value and variance of the terminal wealth are $\mathbb{E}[x_T^*] = \gamma_0 x_0 + \omega \left(\frac{1}{\hat{\theta}_0} - 1 \right)$ and $\text{Var}[x_T^*] = \omega^2 \left(\frac{1}{\hat{\theta}_0} - 1 \right)$, respectively, and the MV efficient frontier can be expressed as $\text{Var}[x_T^*] = \frac{\hat{\theta}_0 (\mathbb{E}[x_T^*] - \gamma_0 x_0)^2}{1 - \hat{\theta}_0}$.

Note that when returns are independent, we only need to solve CCQO problem $\mathcal{P}_{\text{miqp}}^{\text{ccqo}}(\hat{\mathbf{D}}_t, \hat{\mathbf{d}}_t, q)$ for $t = 0, \dots, T-1$. If $q = n$, problem $\mathcal{P}_{\text{cmmv}}^1(\omega, n)$ becomes the conventional MMV portfolio optimization model and the policy (44) generates the policy given in Li and Ng (2000).

4. Numerical example

In this section, we conduct numerical experiments to evaluate the CMMV portfolio optimization model by comparing it with benchmark models. Specifically, we adopt a similar market model in Zhang (2001) and Costa and Araujo (2008). The risky asset pool comprises $n = 10$ assets selected from the Dow Jones

Index, namely IBM (I), Boeing (B), Altria Group (A), Exxon Mobil (E), 3M (M), AIG (G), UTC (U), Johnson & Johnson (J), Procter & Gamble (P), and Caterpillar (C). The initial wealth is $x_0 = 1$ and the investment is quarterly based with $T = 8$ quarters. The statistical parameters of these assets follow the Markov regime-switching model with two major trends in the market, up and down ($s_t \in \mathbb{M} = \{0, 1\}$). For each market state s_t , the quarterly mean returns and covariance matrices of the assets returns $\mathbf{R}_t$ are calibrated in Costa and Araujo (2008). The mean returns are specified as $\boldsymbol{\mu}_t(i) = \mathbf{e} + 10^{-2}\tilde{\boldsymbol{\mu}}(i)$ for $t = 0, \dots, T-1$, where $\tilde{\boldsymbol{\mu}}(i)$ is the one give in Section 6 in Costa and Araujo (2008).⁷ The covariance matrix of the returns are as same as the one in Section 6 in Costa and Araujo (2008). Based on the data of $\boldsymbol{\mu}(i)$ and $\boldsymbol{\Sigma}(i)$, we can compute $\mathbf{d}(i)$ and $\mathbf{D}(i)$ by (8) and (9), respectively. Using the sample (year 2000–2003) of the return, the transition probabilities are estimated as $p(1, 1) = 0.55$, $p(1, 2) = 0.45$, $p(2, 1) = 0.55$, $p(2, 2) = 0.45$.

For comparative purposes, we consider two benchmark models, namely, the static cardinality-constrained MV portfolio (abbreviated as CMV-static) model and the CMMV model with the independent returns (abbreviated as CMMV-independent), where the details are as follows:

- **Benchmark CMV-static:** The investor regards the whole T time periods as one time period and solves the static CMV portfolio optimization (see, e.g., Gao and Li (2013) and Bertsimas and Cory-Wright (2022)), where the covariance matrix and expected return are re-scaled according to the T -period setting. After solving the static portfolio, the investor holds the risky asset until the end of the T period.
- **Benchmark CMMV-independent:** The investor assumes the assets' returns are independent over time (i.e., Assumption 10) and applies the CMMV portfolio for the independent returns given in (44).

In our experiment, we assume that the true assets' returns follow the Markov chain model. Specifically, we generate 10^7 sample paths of the assets' returns $\mathbf{R}_t$ according to the Markov chain model (Assumption 1) by simulating the process of the market state s_t , where $t = 0, \dots, T-1$, based on transition probabilities. For each s_t , we generate the return $\mathbf{R}_t$ according to a multivariate normal distribution $\mathcal{N}(\boldsymbol{\mu}(s_t), \boldsymbol{\Sigma}(s_t))$. Given these sample paths, the two benchmarks adopt different methods to estimate the model parameters.⁸ Once we obtain the portfolio policies for the two benchmark portfolios, along with the CMMV portfolio, we implement these policies on the previous sample paths of the assets' returns and record the terminal wealth of each sample path. In all these cases, the CCQO problems are solved by Gurobi Optimization (2023).

We begin by comparing the trading behavior of different models with the cardinality set at $U(q = 5)$. Fig. 1 depicts typical sample paths of portfolio holding shares for the second asset (where the holding share is defined as $u_t(2)/S_t(2)$ at time t) and the corresponding wealth trajectories in response to market state transitions. The middle sub-figure of Fig. 1 illustrates that the CMV-static policy results in constant holding (indicated by the dotted line) throughout the entire investment

⁷ The meaning of value $\boldsymbol{\mu}_t(i)$ defined in our paper is different from the one in Costa and Araujo (2008). They use $\mu_i(t)\%$ as the return rate under the state i where we use $\boldsymbol{\mu}_t(i)$ to denote the total return. Thus, in our paper, $\tilde{\boldsymbol{\mu}}(i)$ is nothing but $\boldsymbol{\mu}_t(i)$ defined in Costa and Araujo (2008).

⁸ In the CMV-static model, the investor estimates the T -period expected returns and covariance matrix using these samples. However, in the CMMV-independent model, the investor is unaware of the Markov structure of the returns and assumes that the returns are independent and identically distributed, following a multivariate normal distribution.

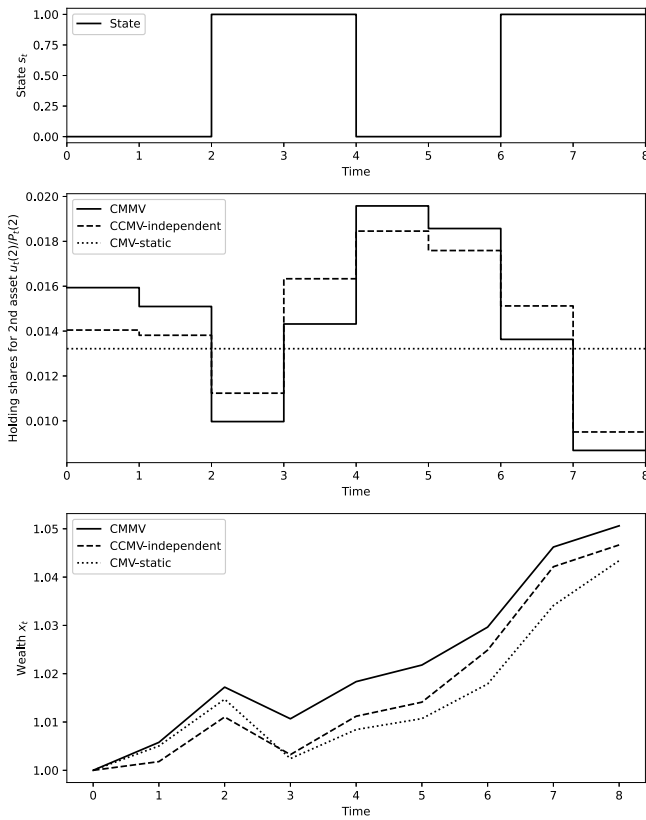


Fig. 1. Comparison of market states, portfolios and wealth sample paths.

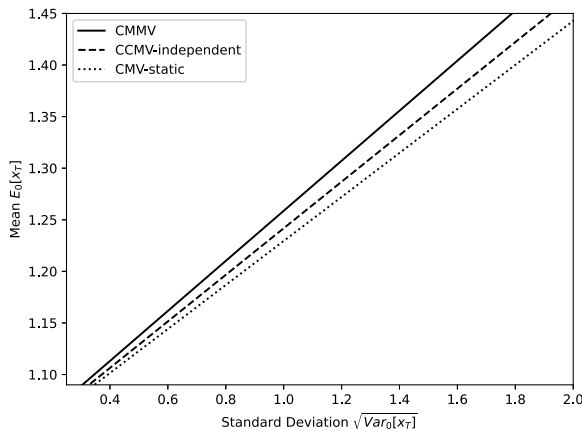


Fig. 2. Comparison of MV efficient frontier for $q = 5$.

horizon. Moreover, our CMMV policy (solid line) demonstrates greater responsiveness to changes in the market state compared to the CMMV-independent policy (dashed line). Specifically, the dynamic CMMV model adjusts its strategy based on prevailing market conditions—favoring a higher stock position during up states and a reduced position during down states, contrasting with the fixed strategy of the CMMV-independent model. Consequently, our dynamic CMMV policy yields significantly higher terminal wealth compared to the other benchmarks, particularly outperforming the static benchmark. These results underscore the benefits of considering dynamic cardinality-constrained MV portfolios under Markov regime-switching market conditions.

Fig. 2 compares the MV efficient frontiers of the terminal wealth generated by our CMMV model with two benchmarks when the cardinality is $q = 5$ and the horizon is $T = 8$. It clearly

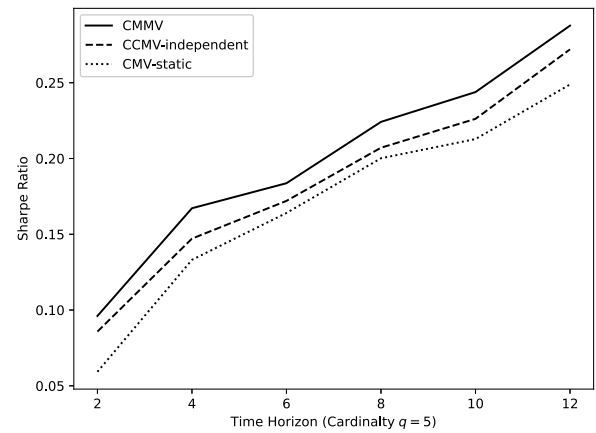


Fig. 3. Comparison of Sharpe Ratio.

shows that the CMMV portfolio outperforms the two benchmark portfolios. The efficient frontier can be equivalently interpreted by the Sharpe ratio, defined in Corollary 7. Fig. 3 reports the horizon effect. As the investment horizon T increases, the Sharpe ratio generated by our CMMV portfolio grows much faster than that of the benchmark portfolios, further demonstrating the superior performance of our CMMV model when the investment horizon is long. This phenomenon is due to the fact that the dynamic feedback policy (28) actively takes advantage of the switching of market states to make portfolio decisions.

We then examine the impact of the cardinality size q on the performance of different portfolio policies. In real market conditions, both management fees and transition costs are significant factors affecting the performance of dynamic portfolio policies. Management fees, typically charged by portfolio managers or financial advisors, represent a fixed cost relative to the investment horizon. Transition costs, which encompass expenses associated with rebalancing or altering a portfolio—such as brokerage fees and taxes—are proportional to transition volumes. Clearly, both of these costs are directly linked to the cardinality parameter q .

We adopt the simple model to capture the impact of these costs as follows. We assume the management fee $M(x_0, q)$ takes a linear function form of q as $M(x_0, q) = \alpha_0 q x_0$ where $\alpha_0 > 0$ is the unit fee and x_0 is the initial wealth. As for the transaction cost at time t , it is charged according to the variation of the asset holding as $TC_t = \alpha_1 \sum_{i=1}^n |u_t(i)/P_t(i) - u_{t-1}(i)/P_{t-1}(i)|$ where α_1 is the unit cost. In the following test, we set $\alpha_0 = 0.1\%$ and $\alpha_1 = 0.05\%$. In our computation, the above costs are deducted from the terminal wealth. Table 1 and Fig. 4 present the performance (the Sharpe Ratio) of different portfolio policies when the above costs are considered. Table 1 reports transaction cost (TC), and Sharpe ratio (SR) as the cardinality size q increases. Initially, the Sharpe ratio increases with cardinality q , which confirms the result in Corollary 7. However, Sharpe ratio decreases when cardinality becomes too high due to the costs. As evident in Fig. 4, a small cardinality results in an insufficiently diversified portfolio, unable to mitigate risks effectively. Conversely, a large cardinality leads to a portfolio with too many assets, resulting in expensive management fees and transition costs. Fig. 4 also shows that, as the management costs and transaction costs are charged in the dynamic portfolio model, CMMV and CMMV-independent models cannot outperform the naive CMV-static model. Therefore, it is crucial to select an appropriate cardinality parameter q to enhance portfolio profitability.

To determine the optimal cardinality q in practice, we propose a data-driven cross-validation approach. This method involves splitting historical data into training and validation sets. For each

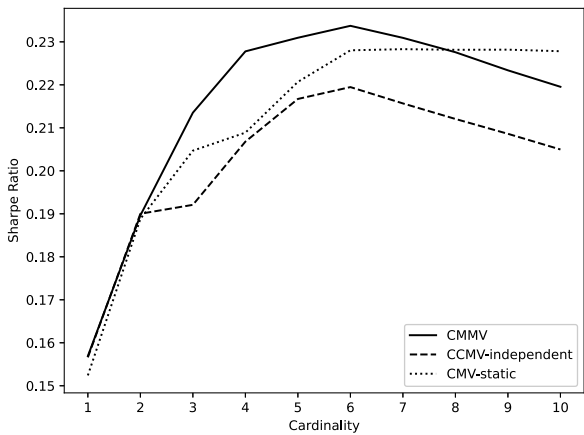


Fig. 4. Sharpe ratios of varying cardinality q with management fee and transition cost.

Table 2
Sharpe ratio (SR) of simulated and empirical data ($q = 5$).

	CMV-static	CMMV-independent	CMMV
Simulated	0.221	0.217	0.231
Empirical	0.151	0.133	0.211

candidate value of q within a reasonable range, we estimate the model parameters using the training data and implement the CMMV portfolio strategy over the validation period. We then calculate the out-of-sample Sharpe ratio, taking into account transaction costs and management fees. The optimal cardinality q^* is selected as the value that maximizes the out-of-sample Sharpe ratio on the validation set. This cross-validation procedure provides a systematic and objective approach to balance the benefits of diversification against the costs associated with managing a larger portfolio, offering a practical guideline for portfolio managers to determine the appropriate cardinality constraint in real-world applications.

Moreover, we conduct backtesting with the real market data. Specifically, using monthly return data of the Dow Jones Industrial Average from January 1995 to December 2024, the market states are classified into bull and bear markets based on a hidden Markov model, and the state transition probability matrix is estimated. Using a rolling window approach, each window employs 10 years (120 months) of historical data for parameter estimation of each asset and tests on the subsequent quarter (3 months), comparing three investment strategies (CMV-static, CMMV-independent, and CMMV).

Table 2 reveals a striking contrast between simulated and empirical performance across all three methods. Notably, the empirical setting presents a more challenging environment than the simulated one, primarily due to parameter estimation inaccuracies and market complexities not fully captured in simulations. While all methods experience performance degradation in the empirical setting, the CMMV approach demonstrates remarkable resilience, with its Sharpe ratio declining only marginally from 0.231 to 0.211 (8.7% reduction). In contrast, the CMV-static and CMMV-independent methods suffer more substantial deterioration, with decreases of 31.7% and 38.7% respectively. This disparity highlights the CMMV method's robustness to parameter uncertainty, as it effectively adapts to changing market regimes through its dynamic feedback mechanism.

5. Conclusion

Due to practical trading (investment) restrictions, investors need to limit their portfolios to a subset of the large asset

pool, known as the cardinality constraint. Although cardinality-constrained mean-variance (MV) portfolio optimization has been studied in a single-period static setting, its extension to dynamic multi-period feedback portfolios is rarely reported in the literature. In this work, we consider the cardinality-constrained multi-period MV (CMMV) portfolio optimization in a market with Markov regime-switching parameters. To address this challenging problem, we show that the embedding scheme (Li & Ng, 2000) also works, enabling us to use dynamic programming to derive the analytical solution. With the help of the special structure of the cardinality-constrained quadratic optimization (CCQO), we can separate the wealth from the value function, allowing us to obtain the solution to the original problem by solving a CCQO problem for each market state at each time period. In this solution process, we recursively compute auxiliary variables $\theta_t(s_t)$ defined for market state s_t at each time t . These variables play a key role in determining the total wealth in risky assets, which is further shown to be linked to the conditional Sharpe Ratio. We also accommodate our solution to some special market settings. Our numerical examples show that the dynamic CMMV policy outperforms the benchmark policies when the market has a regime-switching structure. We examine how the cardinality size affects overall performance when transaction costs and management fees are considered. Although the current work obtains a closed-form solution for the CMMV problem, the results are limited to Markov regime-switching-based markets. In the future, it would be interesting but challenging to study the CMMV problem under more general assumptions of asset returns' correlation structure, such as time-series models.

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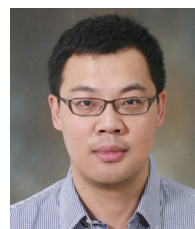


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